CASE STUDY 3

LOGISTIC REGRESSION ON THE TITANIC DATASET

Subject: Fundamentals of Data Science

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Submitted as part of the coursework for Fundamentals of Data Science.

1. Problem Statement

The objective is to develop a logistic regression classifier to predict whether a passenger survived the Titanic disaster. The target variable is **Survived**, where 0 means did not survive and 1 means survived. The task is supervised binary classification.

2. Dataset Overview

The supplied dataset contains **891 rows** and **12 columns**. The target contains 549 non-survivors and 342 survivors. Missing values include Age: 177 (19.9%), Cabin: 687 (77.1%), and Embarked: 2 (0.2%). Survived has no missing values. Cabin, Name, Ticket, and PassengerId were not used as baseline predictors.

Selected features: Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked.

3. Methodology and Preprocessing

The analysis was performed in Python using pandas, NumPy, Matplotlib, Seaborn, and scikit-learn. Exploratory analysis included survival counts, comparisons by sex and passenger class, age and fare distributions, and a correlation matrix. The data was divided into 80% training and 20% testing subsets using a stratified split and random_state=42.

Numerical variables were imputed using the median and standardized. Categorical variables were imputed using the most frequent category and one-hot encoded. The transformations were placed inside a scikit-learn pipeline, so the preprocessing parameters were learned from training data rather than the test set. Logistic regression was fitted with max_iter=2000.

4. Statistical Findings from EDA

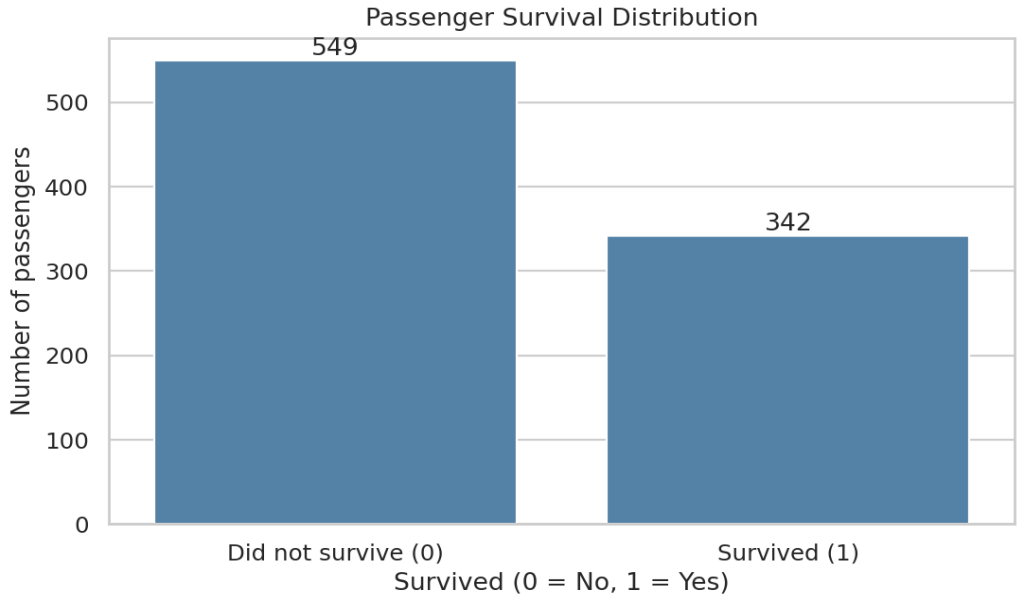


Figure 1. Survival distribution. The dataset has more non-survivors (549) than survivors (342), so accuracy should be interpreted alongside other metrics.

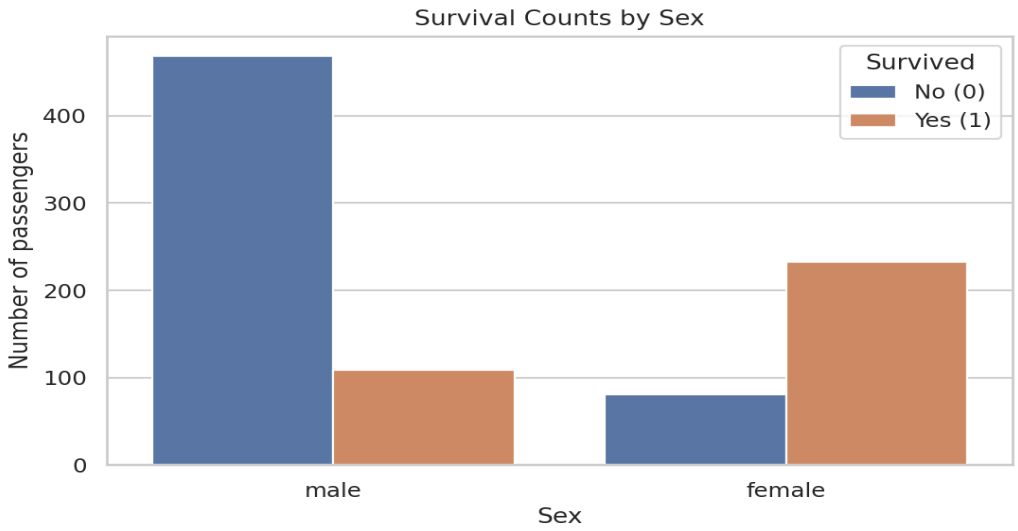


Figure 2. Survival counts by sex. The observed counts and within-group patterns differ between male and female passengers. These are descriptive associations and do not establish causation.



Figure 3. Survival counts by passenger class. Survival patterns vary across classes, making passenger class a potentially informative predictor.

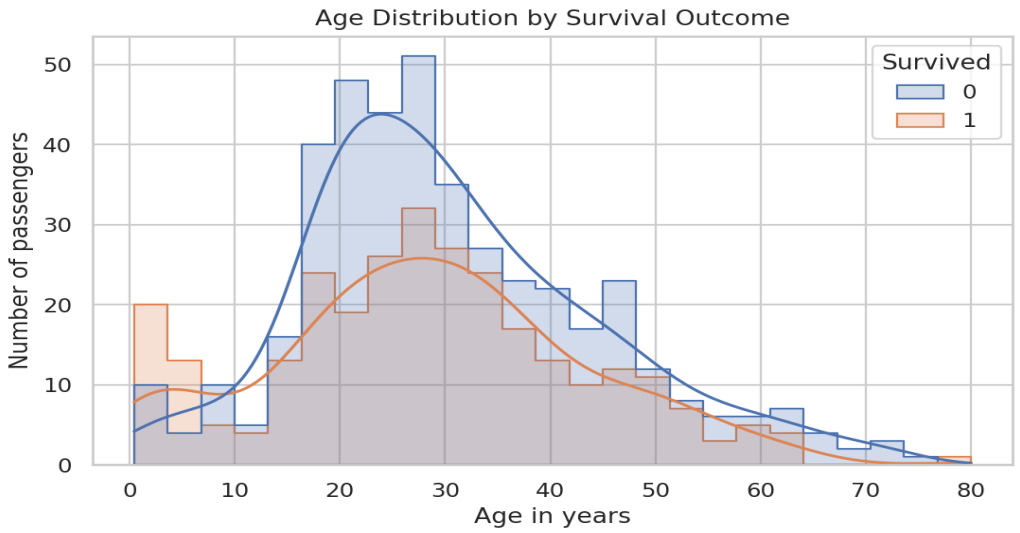


Figure 4. Age distributions by survival outcome. The distributions overlap, so age alone cannot reliably separate the classes.

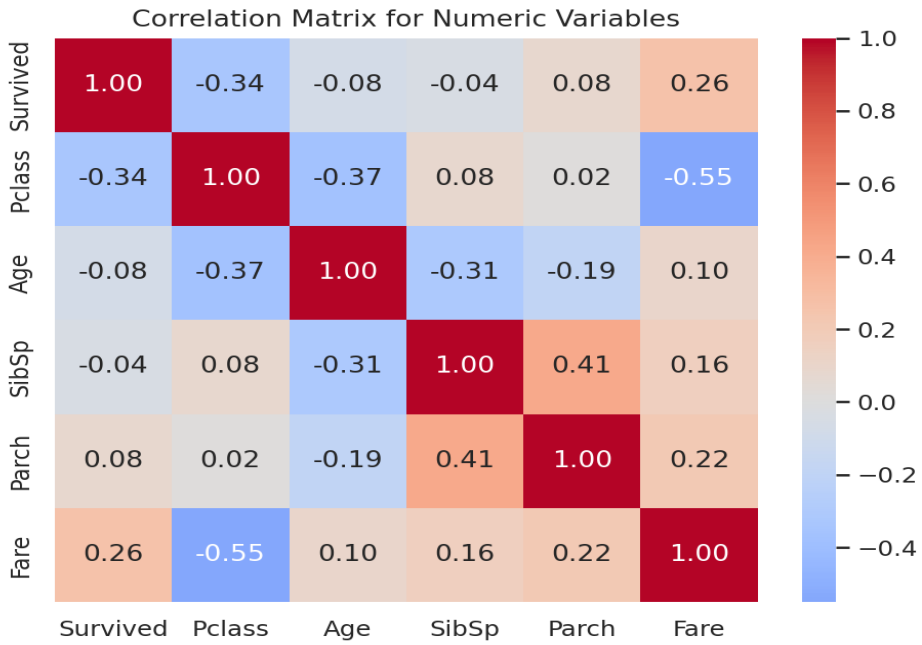


Figure 5. Correlation matrix for selected numeric variables. Correlation captures linear association and does not imply causation. The numeric coding of passenger class should be kept in mind.

5. Model Evaluation

The following metrics were computed from predictions on the held-out 20% test set. Positive class = survived (1).

Metric	Test score	Meaning
Accuracy	0.8045	Fraction of all test predictions that are correct.
Precision	0.7931	Of passengers predicted to survive, fraction who survived.
Recall	0.6667	Of actual survivors, fraction identified by the model.
F1-score	0.7244	Harmonic mean of precision and recall.
ROC-AUC	0.8437	Ability to rank survivors above non-survivors across thresholds.

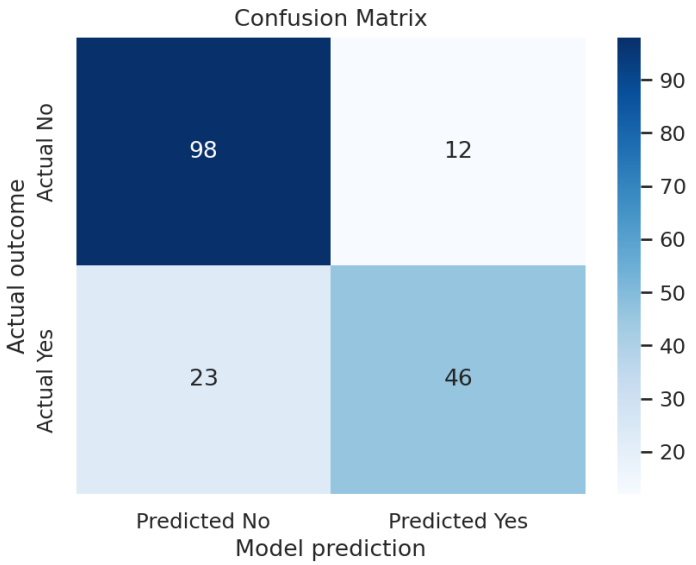


Figure 6. Confusion matrix. Diagonal cells are correct predictions; off-diagonal cells are errors. False negatives are actual survivors predicted not to survive, and false positives are non-survivors predicted to survive.

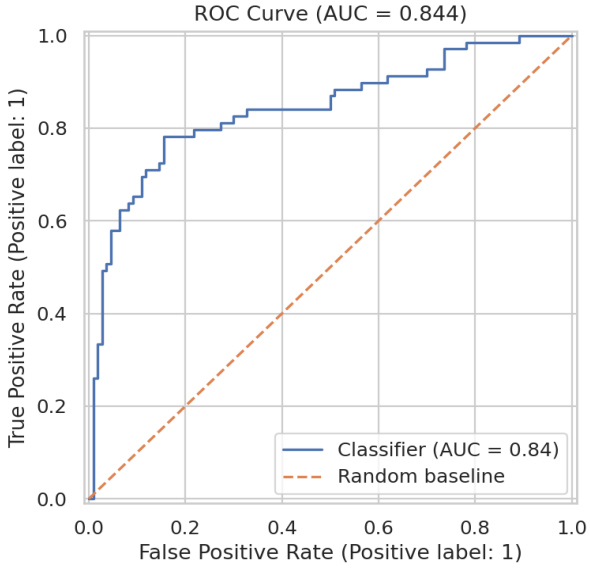


Figure 7. ROC curve. The observed ROC-AUC is 0.844. AUC measures ranking ability and is not the same as accuracy.

6. Interpretation of Model Coefficients

Logistic regression models the log-odds of survival. A positive coefficient is associated with higher predicted log-odds of survival, while a negative coefficient is associated with lower predicted log-odds, holding the other included features constant. Exponentiating a coefficient gives an odds multiplier. Numerical features were standardized, so a one-unit change corresponds to one standard deviation for numerical inputs. One-hot encoded category coefficients are relative to the reference category omitted by the encoder.

cat__Sex_female: coefficient 1.299. This is associated with higher predicted log-odds relative to the relevant reference or standardized numeric scale.

cat__Embarked_Q: coefficient 0.305. This is associated with higher predicted log-odds relative to the relevant reference or standardized numeric scale.

num__Fare: coefficient 0.100. This is associated with higher predicted log-odds relative to the relevant reference or standardized numeric scale.

num__Age: coefficient -0.503. This is associated with lower predicted log-odds relative to the relevant reference or standardized numeric scale.

num__Pclass: coefficient -0.922. This is associated with lower predicted log-odds relative to the relevant reference or standardized numeric scale.

cat__Sex_male: coefficient -1.321. This is associated with lower predicted log-odds relative to the relevant reference or standardized numeric scale.

7. Conclusion

The logistic regression model was trained to predict survival from passenger class, sex, age, family counts, fare, and embarkation port. On the held-out test data, it achieved accuracy 0.804, precision 0.793, recall 0.667, F1-score 0.724, and ROC-AUC 0.844. EDA showed differences in survival patterns by sex and passenger class. These results demonstrate a reproducible classification workflow, but individual predictions are uncertain and the model reflects limitations and biases in the historical dataset. Future work could use cross-validation, tune regularization, and compare other classifiers.

8. Optional Streamlit Deployment

A simple Streamlit application is included in app.py. It loads the saved preprocessing-and-model pipeline, accepts passenger attributes, and displays the estimated survival probability and binary prediction. To run locally: install requirements, then execute `streamlit run app.py` from the project folder. Online deployment is optional under the assignment brief.

Reproducibility and academic integrity

The notebook contains executable code, outputs, visualizations, and interpretations. Before submission, rerun all cells, check that the figures and scores match your own execution, and make sure you can explain each step. Acknowledge the supplied dataset and the Python libraries used. This report should be treated as a learning draft that you verify and personalize, not as a substitute for understanding the analysis.